Dehydration or co-ion pairing: what sets lithium selectivity in near-saturated brines?

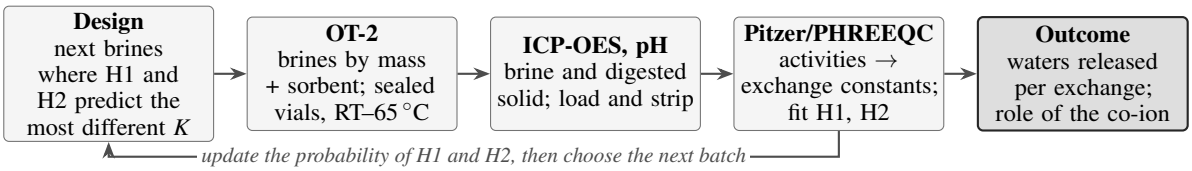
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Research question. In hypersaline lake brines and oilfield produced waters, lithium is a minor ion, outnumbered by Na^+ and Mg^{2+} by one to three orders of magnitude. Some sorbents still take it up selectively. Why, and why does that selectivity change from one brine to the next?

Knowledge gap. Selectivity is usually reported as a distribution coefficient in a few brines, where activities, ion pairing and water activity change at once, so mechanisms rest on correlations and selectivity in a new brine cannot be predicted. Yet for the major ions of natural brines (Na–K–Mg–Ca–Cl–SO_4), Pitzer models give reliable activities up to saturation [1], and binary lithium-salt parameters are tabulated [4]. The solution side can therefore be largely accounted for, and the selectivity that remains belongs to the sorbent.

Hypotheses. *H1, dehydration control:* after Pitzer activity correction, the exchange constant K for Li^+ against Mg^{2+}, at matched solid loading, depends on brine composition only through the water activity a_w, and the slope of $\ln K$ against $\ln a_w$ counts the net waters released per exchange (an osmotic-stress analysis), as expected if Mg^{2+} is excluded by the cost of shedding its hydration shell (-1830 kJ mol^{-1}, against -950 for the two Li^+ that replace it [2]). *H2, cooperative co-ion uptake:* at fixed a_w, selectivity also depends on which anion enters the solid with Li^+. H2 is expected for a layered double hydroxide, which takes up LiCl as a pair [3], but not for a titanate ion sieve, which exchanges Li^+ for H^+, so each sorbent is a control for the other.

Approach. An Opentrons OT-2 prepares synthetic brines by mass in sealed vials: dilution series of a Great Salt Lake composition that take a_w from about 0.98 to 0.75 at fixed ion ratios, and swaps (Cl^- for SO_4^{2-}, Na^+ for Mg^{2+}) designed in PHREEQC [4] to hold a_w constant while staying undersaturated in mirabilite and epsomite. Vials equilibrate with each sorbent at room temperature and on a heater-shaker at 40–65 °C. ICP-OES, on gravimetric, matrix-matched dilutions, measures the brine and the rinsed, digested solid, so Na and Mg loadings are measured rather than taken by difference; pH is logged, since the titanate releases H^+. Equilibrium is confirmed by time series and by approach from both sides (loading and stripping). Bayesian experimental design picks each next batch where H1 and H2 predict the most different K: AI used to tell mechanisms apart, not to optimize uptake.



Expected outcomes. (1) Activity-based Li^+/Na^+ and Li^+/Mg^{2+} exchange constants, with enthalpies from their temperature dependence, for two sorbents with different uptake chemistry; (2) the number of waters released per exchange, and whether co-ion pairing adds to it; (3) a test of whether those constants predict selectivity in real Great Salt Lake and produced-water samples. The result is a mechanistic basis for predicting selectivity in brines not yet measured, a step toward domestic lithium from brines.

Fit and resources. This addresses *Overcoming Thermodynamics via Cooperative Chemical Interactions*: isolating and releasing (loading and stripping) a dilute critical mineral from a complex mixture, where H2 is the cooperative mechanism, with AI used for mechanistic insight. No sorbent screening, optimization or process engineering is proposed. The PI works on self-driving labs and Bayesian experimental design; the lab's OT-2 runs precipitation series checked against PHREEQC/Pitzer; campus ICP-OES and XRD are available.

References. [1] C. E. Harvie, N. Møller, J. H. Weare, *Geochim. Cosmochim. Acta* 48, 723 (1984), doi:10.1016/0016-7037(84)90098-X. [2] Y. Marcus, *J. Chem. Soc. Faraday Trans.* 87, 2995 (1991), doi:10.1039/FT9918702995. [3] M. P. Paranthaman *et al.*, *Environ. Sci. Technol.* 51, 13481 (2017), doi:10.1021/acs.est.7b03464. [4] D. L. Parkhurst, C. A. J. Appelo, USGS Techniques and Methods 6-A43 (2013), doi:10.3133/tm6A43.